Module 32.08: Isolation Forest for Anomaly Detection (Hinglish)

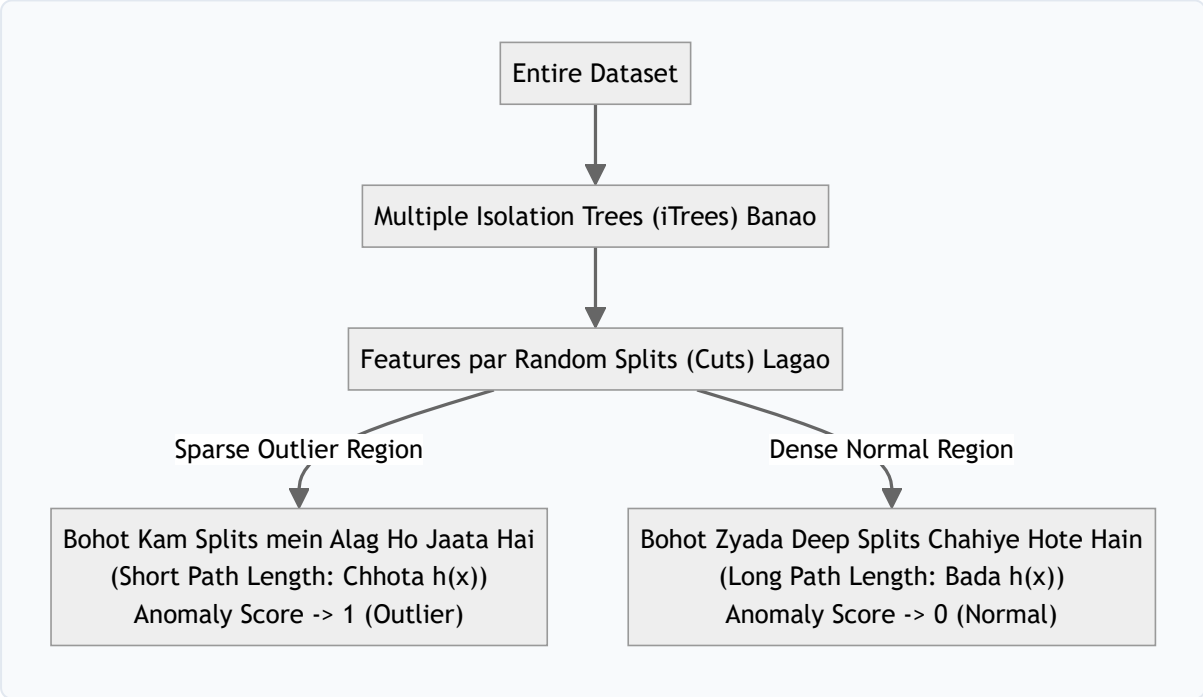
1. Overview & Core Philosophy

Isolation Forest (iForest) unsupervised anomaly detection ke liye ek bohot powerful ensemble tree-based algorithm hai. Baaki algorithms (jaise K-Means ya DBSCAN) pehle normal data ka pattern samajhne ki koshish karti hain aur fir jo bacha use outlier bolti hain. Lekin Isolation Forest **seedhe anomalies ko isolate (alag) karne** par kaam karta hai.

Isolation Forest Ka Basic Rule:

Anomalies do tarah se pehchani jaati hain:

1. **Ginti mein kam (Few)**: Poore data mein sirf $1 - 5\%$ hi anomalies hoti hain.
2. **Features mein bilkul alag (Different)**: Yeh normal dense cluster se dur khali (sparse) jagah par maujood hoti hain.



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2. Isolation Trees (iTrees) Kaise Kaam Karti Hain?

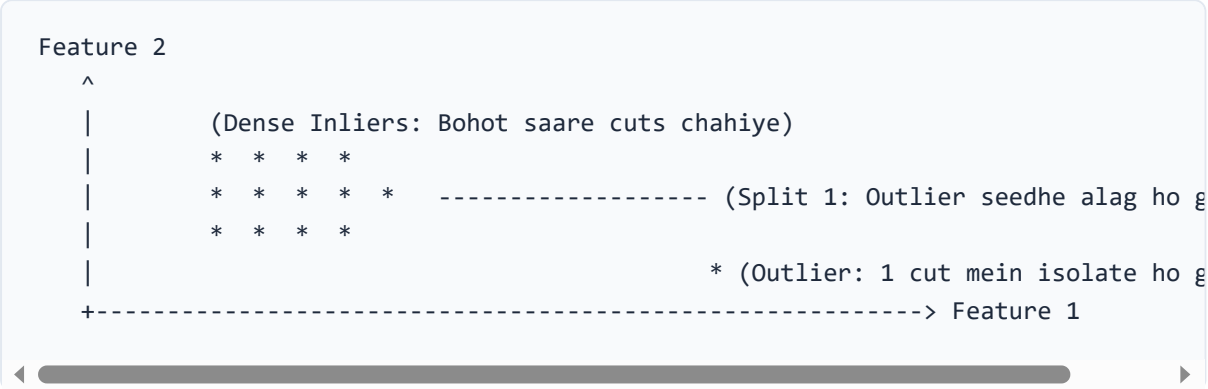
Ek **Isolation Tree** ek binary decision tree hoti hai jisme splits randomly lagaye jaate hain:

1. **Random Feature Selection**: Randomly koi bhi ek feature q select kiya jaata hai.

2. **Random Split Value**: Us feature ki minimum aur maximum value ke beech randomly ek split point p choose kiya jaata hai.
3. **Partition**: Data do hisson mein divide ho jaata hai ($x_q < p$).
4. **Stopping Condition**: Yeh cutting tab tak chalti rehti hai jab tak node mein sirf **ek single sample** na reh jaaye (leaf node ban jaaye).

Outlier Jaldi Kyun Isolate Hote Hain?

- **Outlier**: Kyunki outlier bilkul dur akela khada hai, isliye pehle ya doosre random cut (split) lagte hi woh baaki data se alag ho jaata hai. Isliye outlier ke liye **path length** $h(x)$ (root se leaf tak ki duri) bohot **chhoti** hoti hai.
- **Normal Inlier**: Normal points ek dense bheed (cluster) mein hote hain. Ek single point ko alag karne ke liye ped ko bohot gehra (deep) split karna padta hai. Isliye inliers ke liye **path length** $h(x)$ bohot **badi** hoti hai.



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3. Anomaly Score Ka Mathematical Formula

Multiple trees mein point x ke average path length ko measure karke hum **Anomaly Score** $s(x, n)$ nikaalte hain:

$$\text{Anomaly Score}(x) = 2^{-\frac{E(h(x))}{c(n)}}$$

Formula Ke Components:

Symbol	Kya Hai	Role
x	Single data sample	Jis point ka outlier score nikaal rahe hain
n	Data size	Dataset mein total samples ka count
$h(x)$	Integer	**Path Length**: Tree ke root se leaf tak kitne edges cross kiye

Symbol	Kya Hai	Role
$E(h(x))$	Float	Saare isolation trees mein point x ka **Average Path Length**
$c(n)$	Normalizing Factor	n nodes wale Binary Search Tree (BST) ka average path length

Normalizing Constant $c(n)$ Ka Formula:

Isolation Tree ka mathematical structure Binary Search Tree jaisa hota hai, isliye expected path length:

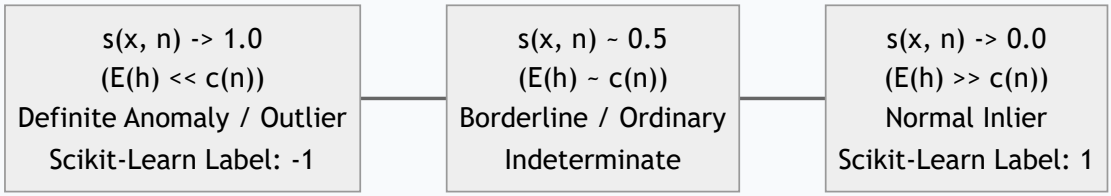
$$c(n) = 2\left(\ln(n-1) + \gamma\right) - \frac{2(n-1)}{n}$$

Yahan $\gamma \approx 0.5772156649$ **Euler-Mascheroni constant** hai.

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4. Anomaly Score Ka Interpretation & Output Labels

Score hamesha 0 aur 1 ke beech rehta hai:



1. **Jab $E(h(x))$ bohot chhota ho ($E(h) \to 0$)**:

$$\frac{E(h(x))}{c(n)} \to 0 \implies s(x,n) \to 2^{-0} = 1$$

   - Point har tree mein turant isolate ho raha hai.
   - **Verdict**: Pakka Outlier / Anomaly $\implies$ **Scikit-Learn Label =** `-1`.
2. **Jab $E(h(x))$ bohot bada ho ($E(h) \to n-1$)**:

$$\frac{E(h(x))}{c(n)} \to \text{large} \implies s(x,n) \to 2^{-\infty} \approx 0$$

- Point ko isolate karne ke liye bohot gehre cuts lagane pade.
- **Verdict**: Normal Inlier $\implies$ **Scikit-Learn Label =** `1`.

3. **Jab** $E(h(x)) \approx c(n)$:

$$s(x,n) \approx 2^{-1} = 0.5$$

- Point bilkul average depth par hai; data mein koi clear anomaly nahi hai.

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5. Isolation Forest Ke Bada Fayde

1. **Koi Distance Matrix Nahi Chahiye**: KNN ya DBSCAN ki tarah har point ka dusre point se distance ($O(N^2 d)$) calculate nahi karna padta.
2. **Super Fast Execution**: Model linear time $O(t \cdot \psi \log \psi)$ mein train ho jaata hai.
3. **High Dimensions Mein Best**: 20 se 100 features wale tabular datasets par bhi yeh bilkul accurate kaam karta hai jahan Euclidean distance fail ho jaata hai.